

WHIMSYLABS · WELCOME PACK

Real practical science, *wherever your students log in.*

Stop clicking “Next.” Start doing science.

What WhimsyLabs does, how it does it, where it differs from the guided-simulation platform you’re currently renewing, and a term-by-term delivery plan for all 52 practicals you sent us.

Prepared for Anthony Peters, King’s InterHigh

ALL 52 MAPPED!

BETT 2025 Kids Judge Award Winner

TechLearning #1 Best of BETT 2026

Global launch September 2026

28 July 2026

We simulate the laboratory, not the lab worksheet

WhimsyLabs is virtual laboratory software for Biology, Chemistry, Physics and Electronics, built on a real physics engine rather than a sequence of animations. Students handle apparatus, take measurements, and get the results their technique actually earned them. It runs in a browser on Chromebooks, Macs and PCs, with VR headset support following closely behind desktop, no headset required, and no installation for web access.



Click to watch the lab in action

See the physics engine, the sandbox and the assessment working together — two minutes.

youtube.com/watch?v=9D2e2e2gzvk

It replaces the bench, and it makes real bench time worth more

Most virtual labs are sold as a substitute for a laboratory. Ours does that, but the second use is the one schools tend to underestimate.



Where there is no bench, it becomes the bench

Students on the UK, Middle East and South-East Asia timetables all run the same practical, at home, on hardware they already own. No travel, no scheduling around a campus, and no cohort excluded by geography. Because the simulation is physics-driven rather than scripted, what they build is genuine procedural skill, and we can evidence it, because we record what they physically did, in what order, and how well.



Where there is a bench, students arrive already competent

When pupils do get into a physical lab, they can arrive having already run the procedure end to end. They know the sequence, the apparatus, the hazards and the PPE. **The health and safety learning is pre-loaded before anyone touches glassware.**

That changes what a lab session can be. Instead of spending it on briefing, demonstration and safety instruction, you start at hands-on method, so the same slot supports something broader, more open-ended, and more genuinely investigative. Students have already made their mistakes somewhere those mistakes were free.

Three systems, one simulation

01 A physicality-first engine

The simulation replicates the chaos and weight of the real world, temperature perturbations, reagent impurities, and deviation between samples. Whether a student is pouring titrant in VR or focusing a microscope on a Chromebook, fine motor control and procedural accuracy are what produce the result.

Drop a beaker and it breaks. Overheat a compound and it reacts. That freedom to fail builds resilience and a genuine understanding of laboratory risk.

Further reading: [Physicality in Virtual Labs](#) · [Why Other Virtual Labs Fail: The Physics Engine Solution](#) · [Sandbox Learning: Why Freedom to Fail Is Essential](#)

02 Assessment that grades the process

We track physical inputs inside the lab, equipment handling, technique, sequence, reaction times, and grade against them. Every student's reagents carry slightly different concentrations and impurities, so every student has a different correct answer.

Follow-up questions are tied to their own data: "why does *your* graph show an unexpected peak?" A chatbot can't answer that, because it doesn't know they skipped rinsing the burette three steps earlier. That is what makes the platform safe to set as unsupervised homework.

Further reading: [OECD: Grade the Process, Not the Product](#) · [Grading the Process, Not the Answer](#) · [94% of Students Use AI for Assessed Work, Not the Problem](#) · [When AI Misreads "At Least One"](#)

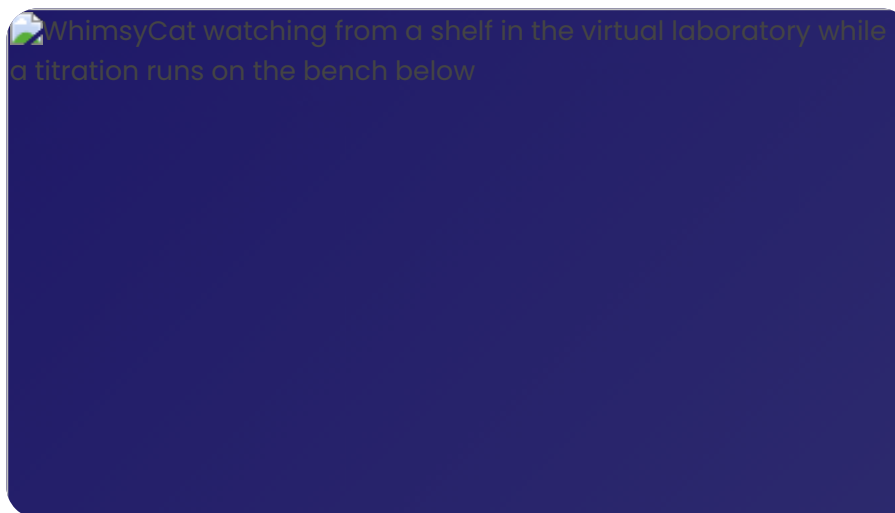
03 WhimsyCat, an AI tutor with no chat window

Our tutor infers a student's understanding entirely from their actions in the lab. Pupils never type a prompt and never receive generated text.

There is no free-text surface to safeguard, no generated content reaching a child, and nothing to prompt-inject. That is a materially easier system to sign off against the DfE's AI safety expectations than a chat-based tutor.

Both the AI grading and WhimsyCat are live from day one of the pilot. Both are also the parts of the platform we most expect to tune in response to what your teachers tell us, the design principles above are settled; the thresholds, prompts and weightings are deliberately not.

Further reading: [How WhimsyCat Meets the DfE AI Safety Standards Without a Chat Box](#) · [WhimsyCat Detects and Responds to Student Frustration](#) · [The Chatbot Reckoning: What 2026's Lawsuits Mean for School AI](#)



WhimsyCat watching from a shelf in the virtual laboratory while a titration runs on the bench below

WhimsyCat watches the bench, not a chat box. It reads what a student does with the apparatus and steps in when the technique goes wrong — there is nowhere for a pupil to type, and nothing generated back at them.

What's in the platform

All four disciplines live in the same simulation, with real interactions between them, an electronics fault behaves like an electronics fault when it's wired into a chemistry rig.

SIMULATION

Open sandbox

Students can combine any reagent with any apparatus to test their own hypothesis, not just the steps a designer anticipated. Curiosity-driven work, safely contained.

SIMULATION

Real consequences

Mistakes propagate the way they do on a real bench. Contaminate a sample and the analysis shows artefacts; add acid too fast and the pH overshoots. Students diagnose their own errors.

SIMULATION

Modelled hazards & PPE

Fully simulated hazard labelling, safety equipment and cleanup. Safety behaviour is observed and scored, not assumed, so pupils who later enter a real lab arrive already briefed.

ASSESSMENT

Automatic practical grading

Skill mastery and safety assessed in real time, with a quick or detailed per-student breakdown across procedure, safety, data collection and scientific communication. Live from day one of the pilot, and tuned through the year on your feedback.

ASSESSMENT

AI-proof by construction

Because every student's data set is unique, answers can't be shared or generated. This is what makes the platform safe to set as unsupervised homework.

TEACHING

Marking time returned

Roughly 3.5 hours per teacher per week recovered from lab setup, observation, cleanup and grading, time that goes back into live teaching.

TEACHING

Intervention dashboard

A real-time view of performance rather than grades alone. Struggling students surface early; reports export, or feed your LMS from individual pupil up to whole-region view.

ROADMAP · 2027/28

AI Experiment Builder

Paste a protocol in plain English and get a working lab scenario in minutes, equipment, reagents, reactions, objectives and assessment criteria. Not in the first year: 2026/27 is dedicated to landing your practicals properly.

ROADMAP · 2027/28

Worldwide Lab Library

Labs built by educators shared across the WhimsyLabs community, browse, adapt, contribute back. Follows the Experiment Builder, since it depends on it.

ACCESS

Runs on what pupils own

96.66% device compatibility, down to basic Chromebooks with 4 GB RAM. Browser-based, with VR an optional enhancement rather than a dependency.

ACCESS

Their headset, your licence

Where a pupil already owns a VR headset, their school licence covers them using it at home for homework and practicals. You get immersive delivery for the students who can take it, without buying hardware. Desktop leads on features; VR follows slightly behind.

ACCESS

Inclusive by default

Full control remapping, text-to-speech, and self-paced modes for SEND learners. A low-bandwidth mode keeps sessions running on unstable home connections.

TEACHING

Support across every time zone

Students on the UK, Middle East and South-East Asia timetables work at different hours. WhimsyCat is available whenever they are, with personalised daily recommendations that keep asynchronous learners on track without a teacher online at 3am.

ASSURANCE

Data protection you can defend

We expect your DPO's questions and we'd rather answer them early than late. Because WhimsyCat has no pupil chat surface, there is no free-text student input to store, review or leak in the first place, the smallest data footprint of any AI tutor category.

ACCESS

LTI integration

Full Learning Tools Interoperability support, including Blackboard and Canvas, so assignment and grade flow sits inside your existing stack.

ENGAGEMENT

Lab Points, not loot boxes

Students earn free points for safety and accuracy, spent on customising their virtual workspace. Intrinsic motivation, with no predatory engagement mechanics.

CURRICULUM

Exam-board aligned

Built against the required practicals mandated by AQA, Edexcel, OCR and Cambridge International, GCSE through A-Level and international equivalents.

CURRICULUM

Bespoke apparatus requests

Pioneer schools get direct access to our development team to request specific apparatus, the mechanism behind the delivery plan later in this pack.

One Fully-Simulated STEM Lab

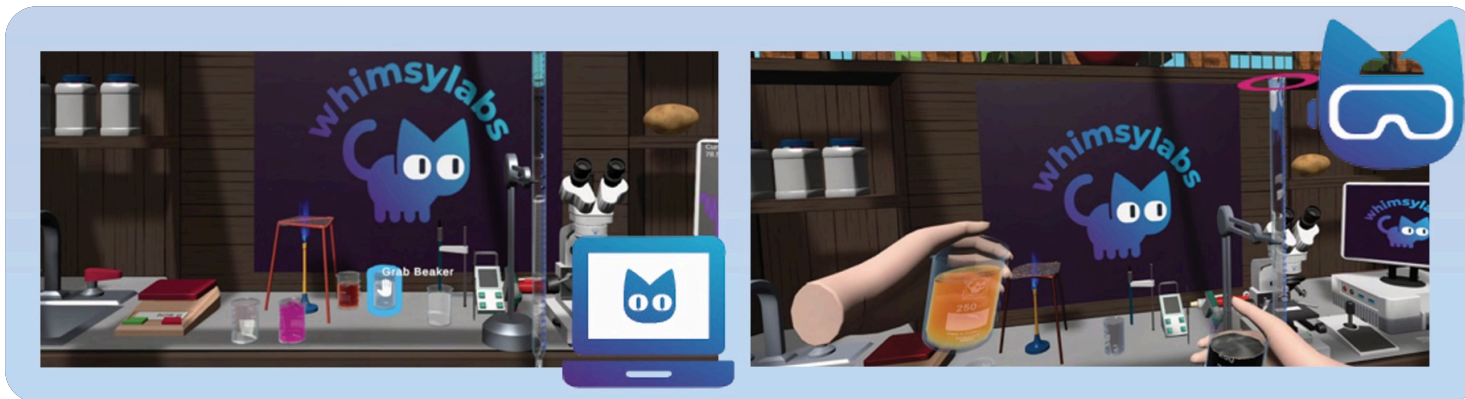
Biology • Chemistry • Physics • Electronics

All disciplines in the same simulation, on VR headsets and desktop, Chromebook, Mac, PC.

Where we differ from guided simulation

The virtual lab market splits into three categories, and the dividing question is simple: **what happens when a student does the experiment wrong?** Video tools show a recording, watching a titration is not performing one. Guided step-by-step simulations, walk students along a predetermined sequence and return them to the correct path when they stray. Physics-engine sandboxes calculate outcomes from physical law, so there is no correct path to be returned to.

If a student can't **make a mess of it**, they aren't doing it.



One sandbox, two ways in. The same bench, the same physics and the same assessment, whether a student is on a Chromebook or in a headset — and in neither case are they following a script.

	GUIDED SIMULATION	WHIMSYLABS
Core interaction	Click through a branching, quiz-gated script.	Manipulate apparatus in an open sandbox governed by a physics engine.
Getting it wrong	Blocked, corrected, or nudged back on rails. The link between action and consequence is broken.	The error happens and propagates. Students must notice it, diagnose it, and decide what to do.
What's assessed	Answers selected and steps completed.	Technique, handling, sequence, safety and cleanup, then the answers.
Homework integrity	Identical data for every student, so answers are shareable and AI-answerable.	Per-student reagent variation means each pupil has a unique correct answer.
AI tutor safety	Typically a pupil-facing chat surface generating text to children.	No student chat window at all. The tutor reads actions, never prompts.
Coverage model	A fixed catalogue, typically a few hundred authored simulations, on the vendor's release schedule.	In 2026/27 we build to <i>your</i> specification, on your term dates, the 52 practicals below. From 2027/28 the Experiment Builder removes the cap entirely.

GUIDED SIMULATION		WHIMSYLABS
Custom practicals	Vendor roadmap request, on their priorities.	Pioneer schools get direct access to our development team to request specific apparatus and reorder the build queue. Self-service authoring arrives in year two.

We'd encourage you to test both platforms against the same practical with the same students, rather than take our word for any of the above. If it helps to have a neutral checklist for that, we wrote one: [Virtual Lab Software for Secondary Schools: A Buyer's Guide](https://whimsylabs.ai/blog/how-to-choose-virtual-lab-software-school/) (<https://whimsylabs.ai/blog/how-to-choose-virtual-lab-software-school/>).

Where guided platforms genuinely win, today. A mature catalogue of several hundred polished, curriculum-tagged simulations is a real asset, and on raw title count we are behind. Our position is that title count is the wrong metric for a school that needs to evidence practical *skill*, but you should weigh that yourself, and the delivery plan below is written so you never have to take it on trust.

Year one builds your practicals. Year two hands you the tools.

We want to be exact about sequencing here, because it's the thing most likely to be misread. **Self-service custom experiments are not part of the 2026/27 offer.** The whole of the first year is committed to building the 52 practicals on your list properly, and we would rather do that than spread the same engineering across two goals and do neither well.

What you get instead in year one is direct access to our development team. If a practical matters more to your timetable than to our sequencing, you tell us and we

move it. That is a slower loop than authoring it yourself, but it is a considerably faster one than a feature request into a vendor backlog.

From **2027/28**, the AI Experiment Builder takes a protocol written in ordinary English and produces a working lab, equipment, reagents, reactions, objectives and assessment criteria. At that point the procurement question changes: you are no longer buying a catalogue and hoping it covers your specification, you are buying an engine plus the ability to author against it. Anything your teachers build can stay private to King's InterHigh, be shared across Inspired, or be contributed to the global library, your call, per lab.

Your 52 practicals, staggered across 2026/27

We will not pretend all 52 will be finished for September, and you should be sceptical of any vendor who says otherwise. Below is an honest four-wave plan aligned to term boundaries.

The sequencing isn't arbitrary, and September is not a sampler. Wave 1 delivers **complete capability families rather than isolated experiments**: the entire titration platform — acid–base, redox, pH curves, K_a determination and Vitamin C — arrives together, as does the whole electrical bench, chromatography across both Biology and Chemistry, and dissection in all its forms. A department can teach a coherent unit from day one rather than working around gaps.

Later waves then inherit those rigs, because once a bench exists the second practical built on it costs a fraction of the first. Wave dates are indicative pending your 2026/27 term boundaries — send us those and we will convert them into dated commitments.

WAVE 1

September launch

Term dates indicative · from Sept 2026

17 practicals

BIOLOGY

- Microscopy & Cell Division
- Organ Dissection (all types)
- Chromatography
- Vitamin C Titration
- Brine Shrimp Development

CHEMISTRY

- Titration (Acid–Base & Redox)
- Neutralisation & pH Curves
- K^a & pK^a Determination
- Chromatography & TLC

PHYSICS

- I–V Characteristics
- Resistance & Resistivity
- EMF & Internal Resistance
- Capacitor Charge/Discharge
- Refraction
- Simple Harmonic Motion
- Viscosity
- Radioactivity (Inverse Square, Absorption)

Complete capability families rather than isolated experiments: the full titration platform, the entire electrical bench, chromatography serving both sciences, and dissection in all its forms. Every rig here is reused by a later wave.

WAVE 2**Autumn term** Term dates indicative · Oct–Dec 2026

12 practicals

BIOLOGY

- Enzymes & Temperature Effects
- Osmosis
- Cell Membranes
- Food Tests (Benedict's, Biuret)

CHEMISTRY

- Making Salts
- Tests for Ions
- Combustion & Enthalpy
- Electrolysis

PHYSICS

- Density
- Acceleration & Freefall
- Hooke's Law & Young Modulus
- Thermal Insulation & Specific/Latent Heat

Core GCSE benchwork across all three sciences. Electrolysis here sets up electrochemical cells in Wave 3, and the tensile rig introduced with Hooke's Law carries plant fibre strength in Wave 4.

WAVE 3**Spring term** Term dates indicative · Jan–Mar 2027

12 practicals

BIOLOGY

- Respiration
- Microbiology & Aseptic Technique
- Heart Rate
- Spirometry

CHEMISTRY

- Reaction Rates & Activation Energy
- Molar Volume of Gas
- Electrochemical Cells
- Organic Analysis (Alcohols, Aldehydes, Carboxylic Acids)

PHYSICS

- Wave Properties & Stationary Waves
- Diffraction
- Speed of Sound
- Gas Laws (Boyle's Law)

A-Level and IB weighting, timed to land before spring coursework deadlines. The wave-optics bench arrives in one block, and the gas rig serves both molar volume and Boyle's Law.

BIOLOGY

- Photosynthesis & Chloroplast Isolation
- Gel Electrophoresis
- Fieldwork & Sampling (Quadrats, Transects)
- Plant Fibres & Tensile Strength
- Habituation Studies

CHEMISTRY

- Organic Synthesis — Solid (Aspirin)
- Organic Synthesis — Liquid (Distillation)
- Hydrolysis Rates (Halogenoalkanes)
- Transition Metal Complexes

PHYSICS

- Magnetic Fields & Forces
- Momentum & Collisions

The specialist builds — field sampling environments, behavioural studies and multi-step organic synthesis — each needing bespoke environment work rather than rig reuse.

The order is not fixed. If a practical matters more to your timetable than to our sequencing, tell us and we will move it, Pioneer schools have direct access to the development team precisely for this, and reordering the queue is the single most useful thing you can do with that access.

Send us your scheme of work alongside your term dates and we will re-cut these four waves against it before anything is signed.

The arguments above, at full length

Everything in this pack is set out in more detail on our blog, with sources. These are the pieces most likely to be useful when you circulate your recommendation.

WHY THE ENGINE MATTERS

Physicality in Virtual Labs: Beyond Traditional Simulations (<https://whimsylabs.ai/blog/physicality-in-virtual-labs/>)

Why Other Virtual Labs Fail: The Physics Engine Solution (<https://whimsylabs.ai/blog/why-traditional-virtual-labs-fail-physics-engine/>)

Sandbox Learning: Why Freedom to Fail Is Essential for STEM (<https://whimsylabs.ai/blog/sandbox-learning-revolution-stem-education/>)

The Critics Are Right: Most EdTech Is Useless (<https://whimsylabs.ai/blog/edtech-critics-right-passive-learning-vs-active-labs/>)

Real-Time Physics: Why Dynamic Simulations Make Learning Fun (<https://whimsylabs.ai/blog/science-real-time-physics-simulations-virtual-labs/>)

ASSESSMENT & AI INTEGRITY

OECD: Grade the Process, Not the Product (<https://whimsylabs.ai/blog/oecd-process-oriented-assessment-validation/>)

Grading the Process, Not the Answer (<https://whimsylabs.ai/blog/process-based-lab-assessment-future/>)

AI Detection Doesn't Work. Process-Based Assessment Does. (<https://whimsylabs.ai/blog/ai-assessment-crisis-solution/>)

AI Boosted Test Scores 127%. Then Students Couldn't Think Without It. (<https://whimsylabs.ai/blog/oecd-ai-learning-paradox-virtual-labs/>)

76% of Teachers Use AI. 66% Say It Erodes Critical Thinking. (<https://whimsylabs.ai/blog/neu-survey-ai-critical-thinking-decline/>)

AI SAFETY & DATA PROTECTION

How WhimsyCat Meets the DfE AI Safety Standards Without a Chat Box (<https://whimsylabs.ai/blog/dfе-ai-safety-standards-tutor-checklist/>)

The Chatbot Reckoning: What 2026's Lawsuits Mean for School AI (<https://whimsylabs.ai/blog/ai-chatbot-lawsuits-2026-schools/>)

Purpose-Built AI vs ChatGPT: Why Design Matters (<https://whimsylabs.ai/blog/purpose-built-ai-education-difference/>)

10 Questions to Ask EdTech Vendors After the PowerSchool Breach (<https://whimsylabs.ai/blog/edtech-vendor-security-questions-powerschool/>)

VR STRATEGY & DEVICES

Your School's VR Headsets Just Lost Their Roadmap (<https://whimsylabs.ai/blog/meta-quest-education-support-cliff-2030/>)

VR Winter? Why Web-First Virtual Labs Are Smart for Schools (<https://whimsylabs.ai/blog/vr-winter-web-first-virtual-labs/>)

VR Headsets vs Chromebooks: A 2026 Cost Guide (<https://whimsylabs.ai/blog/vr-headsets-vs-chromebooks-cost-per-student/>)

New VR Research: Technology Without Pedagogy Falls Flat (<https://whimsylabs.ai/blog/vr-stem-education-research-pedagogical-scaffolding/>)

VR Prepares Students for STEM Careers with Physical Lab Skills (<https://whimsylabs.ai/blog/virtual-reality-prepares-students-real-world-stem-careers/>)

INCLUSION & SEND

Virtual Kidney Dissection Enhances SEND Student Engagement (<https://whimsylabs.ai/blog/virtual-kidney-dissection-send-engagement/>)

What the 2026 SEND White Paper Means for Science Teaching (<https://whimsylabs.ai/blog/send-white-paper-2026-science-practicals/>)

How to Evidence Inclusion in Science for Ofsted's New Toolkit (<https://whimsylabs.ai/blog/ofsted-inclusion-2026-science-evidence/>)

CURRICULUM, COST & POLICY

Triple Science for All by 2028: Delivering the Practicals (<https://whimsylabs.ai/blog/triple-science-entitlement-2028-virtual-labs/>)

Virtual vs Physical Labs: A Cost-Benefit Analysis (<https://whimsylabs.ai/blog/virtual-labs-vs-physical-labs-cost-benefit-analysis/>)

Virtual Labs and the STEM Teacher Shortage (<https://whimsylabs.ai/blog/virtual-labs-solve-stem-teacher-shortage-crisis/>)

Virtual Lab Software for Secondary Schools: A Buyer's Guide (<https://whimsylabs.ai/blog/how-to-choose-virtual-lab-software-school/>)

Green Labs: Sustainability in STEM (<https://whimsylabs.ai/blog/green-labs-sustainability-virtual-stem-education/>)

BY DISCIPLINE

Virtual Biology Lab: Dissections, Microscopy and More (<https://whimsylabs.ai/blog/virtual-biology-lab-dissections-microscopy/>)

Virtual Chemistry Lab: A Teacher's Complete Guide (<https://whimsylabs.ai/blog/virtual-chemistry-lab-teachers-guide/>)

Virtual Physics Lab: Simulations That Actually Teach (<https://whimsylabs.ai/blog/virtual-physics-lab-simulations-teach/>)

ABOUT US

WhimsyLabs Wins Tech&Learning #1 Best of BETT 2026 (<https://whimsylabs.ai/blog/whimsylabs-wins-techlearning-best-of-bett-2026/>)

The WhimsyLabs Story: From Startup to BETT 2025 Winner (<https://whimsylabs.ai/blog/whimsylabs-education-revolution/>)

Assessment in the Age of AI: Our Pearson Webinar (<https://whimsylabs.ai/blog/pearson-webinar-vr-assessment-ai-age/>)

Teachers Are the Experts. We Just Build the Tools. (<https://whimsylabs.ai/blog/teachers-are-experts-custom-experiment-designer/>)

Getting to a September decision

Five things between here and a signed pilot. None of them require you to win an internal argument on a renewal deadline.

01 **Confirm term dates and cohort scope**

We need your 2026/27 term boundaries to convert the waves above into dated commitments, plus which cohorts are in scope, KS3, IGCSE, A-Level, IB, and approximate student numbers.

02 **Technical check on home devices**

Your students sit across UK, Middle East and South-East Asia timetables on their own hardware. We'd like a short call with your IT lead to confirm the device floor and bandwidth profile before anything is committed.

03 **Run a term in parallel, not a rip-and-replace**

We would rather sit alongside your incumbent for the autumn term on one or two cohorts, with agreed success measures, than ask you to displace them on a payment deadline. Compare both on the same practical, with the same students.

04 **Pricing and Pioneer terms**

Pioneer schools secure the 2026/27 academic year at our introductory BETT rate, with beta AI grading and early modules available from the summer term. We'll issue a written quote once scope and cohort numbers are settled.

05 The wider Inspired conversation

If King's InterHigh goes well we're glad to talk to the wider group, but we'd rather earn that on evidence from your cohorts than price for 125 schools before we've served one properly.

Become a Pioneer School